NEONATAL LIFE SUPPORT ALGORITHM

GUIDELINES 2025
EUROPEAN RESUSCITATION COUNCIL©

Antenatal counselling
Team briefing and equipment check

Birth
Start the clock

Delayed cord clamping
Gentle stimulation
Dry, wrap, hat
Keep warm

Assess breathing & heart rate

Inadequate breathing:
Open airway

If apnoea or gasping:
Give 5 inflations
Start with **30 cm H_2O, PEEP 6 cm H_2O, 21% O_2**
Consider SpO_2 +/- ECG

Reassess

If no increase in heart rate:
Chest movement?

YES

If chest is **not** moving:
- Check mask, head position, jaw thrust
- Try other airway opening techniques
- Consider higher inflation pressure

Repeat 5 inflations

Reassess

NO

If no increase in heart rate:
Chest movement?

YES

Continue PPV
Reassess after 30 s

If heart rate remains <60 min^{-1}:
Start chest compressions
- 3 compressions : 1 ventilation
- 100% O_2
- Consider SGA or tracheal tube

Reassess every 30 s

If heart rate remains <60 min^{-1}:
- Vascular access (UVC/IO)
- Consider drugs & fluids
- Consider other factors

Update parents
Debrief team

e.g.
- Pneumothorax
- Hypovolaemia
- Congenital abnormality
- Equipment failure

- Adrenaline: 10-30 µg kg^{-1} every 4 mins
- Intravascular volume (blood / fluid): 10 mL kg^{-1}
- Glucose 10% (if blood glucose is low): 2 mL kg^{-1}

Preterm < 32 weeks

Place undried body into plastic bag, hat
Use external heat source

If breathing: support with **CPAP 6 cm H_2O, O_2 ≥ 30%**

Start with **25 cm H_2O,** O_2 ≥ 30%

Target SpO_2
Right hand
3 min 70 - 75%
5 min 80 - 85%
10 min 85 - 95%

Titrate oxygen to achieve target saturations

Is more help needed?

Maintain normal temperature

CPAP - continuous positive airway pressure; ECG - electrocardiogram; IO - intraosseous; PEEP - positive end-expiratory pressure; PPV - positive pressure ventilation; SGA - supraglottic airway; SpO_2 - oxygen saturation; UVC - umbilical venous catheter